

Summary: Your report presents a well-structured and insightful exploration of face verification technology using Autoencoders (AEs) and Variational Autoencoders (VAEs). The introduction effectively sets the stage, though it could benefit from a more nuanced discussion of AEs and VAEs. The background and related work sections are comprehensive, but streamlining the content to avoid redundancy would enhance clarity. The methodology is detailed and well-organized, with room for further development on VAE training specifics for the target task. Overall, your work demonstrates a solid understanding of the subject but some corrections need to be taken into account for the next stages. Please see the below details:

The Introduction is good but be careful about how you introduce VAE and AE, these approaches are not this popular, due mostly to underperforming. The papers available do not evaluate the performances on many evaluation datasets as well. Which makes the literature on the subject quite weak. This is why it is interesting to study them further!

Try not to repeat yourself between the background and the related work section. It is a little bit challenging to create two separate sections, but in this case, in the background section, I would remove the mention of traditional approaches, hidden markov models and only focus on CNN, since it is the architecture you will compare yourself with. Also, you should develop in the related work, what has already been done with AE and VAE.

In the background section you can mention the benchmark datasets that can evaluate the models in these different scenarios (pose variation, age variation etc)

The methodology part is good, but for the next step, keep in mind you need to develop more about VAE. For now you are introducing the generic VAE, which will have its intrinsic limitations. So the next questions will be: How to train a VAE so that the mean prediction of the encoder learns to project images of the same person at (approximately) the same position?

For the Resnet18 results, there is still a performance issue, highlighted by the distributions you plotted with the Arcface Loss. I will probably provide you the results I obtain with my code, which achieve more than 92% of accuracy on LFW in one epoch, but it would be good if you can solve the issue on your side.

Finally, it is really important to rewrite the captions so that they are more explicit. It is sometimes (especially for the results reports) to understand what you are talking about (What evaluation set is taken into account, which architecture, etc...). See for instance figures 5.6 and 5.5. But overall, they all need to be more complete: Figure 5.7 would give something like "Distance distributions obtained with a Resnet 18 trained with the CrossEntropy loss on CASIA, obtained on LFW evaluation set".

Grade : 16/20

Your report demonstrates a solid understanding of the subject matter and presents a well-organized approach to your research. With some refinements in the introduction of AEs and VAEs, clearer distinctions between background and related work, and more explicit figure captions, your work will be even more impactful. For the next stages, you should focus on getting results with the VAE/AE approach.